Examiner only

4. In 1825, plant collectors brought Japanese knotweed *(Fallopia japonica)* into the UK. It spread into many habitats, mostly near rivers. It is now out of control in most areas, eliminating other plant species and damaging roads and buildings.

Japanese knotweed in summer

sap-sucking louse

Japanese knotweed grows rapidly in summer. Plants reach 4 metres in height and underground stems grow to 25 metres in length.

Scientists working for the Welsh government investigated the use of a sap-sucking louse *(Aphalara itadori)*, to destroy Japanese knotweed in a number of trials in parts of the UK.

In the trials, the louse reduced the growth of Japanese knotweed by 60%. The louse did not harm any other species and reproduced quickly in summer. Most of the lice, however, died in the winter.

(a) (i) Which of the following describes Japanese knotweed in the UK? Write the correct letter in the box. [1]

A an endangered alien species

B an alien invasive species

C an endangered native species

D a native invasive species

answer ☐

(ii) How does Japanese knotweed affect biodiversity in the areas where it grows in the UK? Give a reason for your answer. [1]

....

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(b) (i) State the scientific term used when an organism is used to destroy a pest species. [1]

....

(ii) Calculate the length of underground stems produced in Japanese knotweed when the sap-sucking louse is present. [2]

length = ... m

(c) (i) Following the trials, the scientists concluded that the sap-sucking louse was effective against Japanese knotweed as it reduced growth by 60%.

They also decided that it would be suitable to use this method on a wider scale throughout the UK. Give **one** piece of evidence which supports this decision. [1]

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(ii) How could these scientists check that the results were reproducible? [1]

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